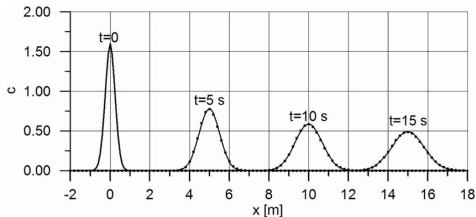


Equations to be solved

$$\frac{\partial \phi(t, x, y, z)}{\partial t} + \nabla \phi(t, x, y, z) \mathbf{u}(t, x, y, z) = D \Delta \phi(t, x, y, z)$$



Initial conditions:

$$\phi(t = 0, x) = \delta(0) \quad \phi(t = 0, x) = 0.5[1 + \sin(2\pi Kx)]$$

Analytical solutions:

$$\phi(t, x) = 0.5 \left[\operatorname{erfc} \left(\frac{x - ut}{2\sqrt{Dt}} \right) + \exp \left(\frac{ux}{D} \right) \operatorname{erfc} \left(\frac{x + ut}{2\sqrt{Dt}} \right) \right]$$

$$\phi(t, x) = 0.5[1 + \exp(-D(2\pi K)^2 t) \sin(2\pi K(x - ut))]$$

LBM algorithm

- Initialize DFs e.g. $f_i^{\text{eq}}(\rho_{\text{ini}}, \mathbf{u}_{\text{ini}})$
- Setup τ according to material properties
- Repeat until steady state or desired time is achieved
 1. Compute macroscopic quantities from f_i
 2. Compute equilibrium DF's and post-collision state
 3. Stream f_i^* along lattice links
 4. Apply BC

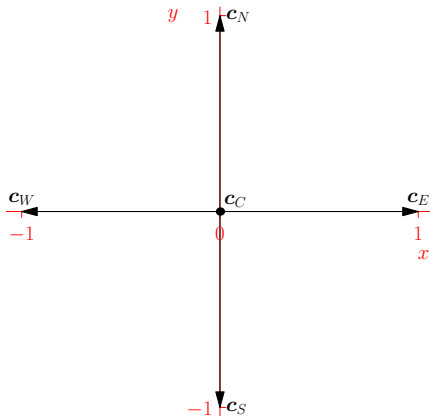
$$f_i(t+1, \mathbf{x} + \mathbf{c}_i) = f_i(t, \mathbf{x}) + \frac{1}{\tau}(f_i^{\text{eq}}(t, \mathbf{x}) - f_i(t, \mathbf{x})) + S_{\phi,i}(t, \mathbf{x}) = f_i^*$$

$$D_{lb} = c_s^2 \left(\tau - \frac{1}{2} \right), \quad f_i^{\text{eq}}(t, \mathbf{x}) = w_i \phi(t, \mathbf{x}) \left(1 + \frac{\mathbf{c}_i \cdot \mathbf{u}(t, \mathbf{x})}{c_s^2} \right)$$

macro-variables...

$$\phi = \sum_i f_i(t, \mathbf{x})$$

LBM D2Q5 Model



	C	E	N	W	S
c_i	(0,0)	(1,0)	(0,1)	(-1,0)	(0,-1)
w_i	1/3	1/6	1/6	1/6	1/6

lattice speed of sound $c_s = \frac{1}{\sqrt{3}}$

Physical & lattice units relations

Advection-diffusion has one dimensionless nr.: Peclet number
(ratio of advection and diffusion scales)

$$Pe = \frac{UL}{D}$$

In LBM, dim-less nrs. must be equal

$$Pe^P = Pe^{LBM} \iff \frac{UL}{D} = \frac{u_{lb}N}{D_{lb}}$$

Mach number is defined as

$$Ma = \frac{U}{c_s} = \frac{u_{lb}}{c_{s,lb}}$$

so we can control time step by Ma and advection-diffusion by Pe numbers ($\Delta x = L/N$)

$$\Delta t = (\Delta x)^2 \frac{D_{lb}}{D} = \frac{u_{lb}^2 D}{u^2 D_{lb}}$$